

University of Batna 2

Faculty of Mathematics and Computer Science

Departement of Mathematics

Chapter 02: Complex numbers

By : Brahimi Mahmoud

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$e^{in\theta} = \cos n\theta + i \sin n\theta$$



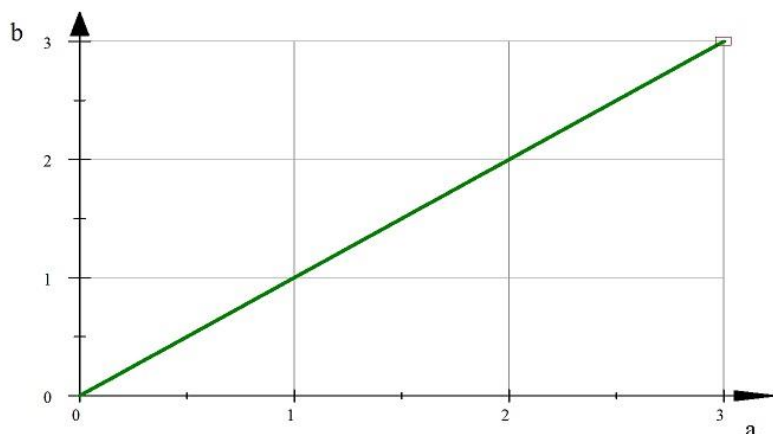
Academic year 2024/2025

Chapter 2 Complex Numbers

Definition

A complex number z is a pair $(a, b) \in \mathbb{R}^2$ which we will denote by $a + ib$, where:

- a is the real part of z , $\text{Re}(z)$
- b is the imaginary part of z , $\text{Im}(z)$
- i is an imaginary number that satisfies the relation $i^2 = -1$.



Complex number $3 + 3i$

Remarks

- We denote by $\mathbb{C}$ the set of complex numbers.
- If $b = 0$, ($\text{Im}(z) = 0$), In this case we will say that z is **real**.
- If $a = 0$, ($\text{Re}(z) = 0$), In this case we will say that z is **pure imaginary**.
- The arithmetic operations in $\mathbb{C}$ are the same as in $\mathbb{R}$, taking into account $i^2 = -1$.

Operations

If $z = a + ib$ and $z' = a' + ib'$ then:

1. $z + z' = (a + a') + i(b + b')$
2. $z - z' = (a - a') + i(b - b')$
3. $zz' = (aa' - bb') + i(ab' + ba')$

Remarks

- $z = z' \Leftrightarrow (a = a') \wedge (b = b')$
- $z = 0 \Leftrightarrow (a = 0) \wedge (b = 0)$
- There is no natural order on $\mathbb{C}$, so one should never write $z \geq 0$ or $z \leq z'$.

Conjugate complex number

Definition

The conjugate of the complex number $z = a + ib$ is the complex number $\bar{z}$, such that $\bar{z} = a - ib$.

Examples:

- $z = 3 - 2i \Rightarrow \bar{z} = 3 + 2i$
- $z = 4 \Rightarrow \bar{z} = 4$
- $z = 3i \Rightarrow \bar{z} = -3i$

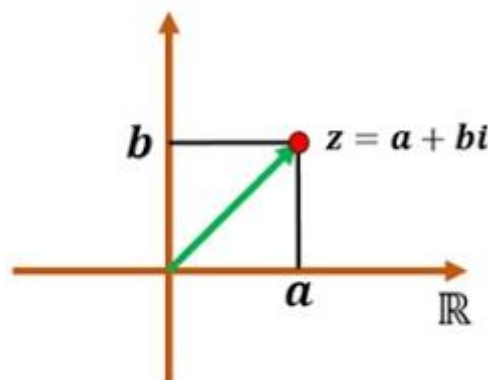
Properties

1. $\overline{z + z'} = \bar{z} + \bar{z'}$
2. $\bar{\bar{z}} = z$
3. $\overline{zz'} = \bar{z}\bar{z'}$
4. $\overline{\left(\frac{1}{z}\right)} = \frac{1}{\bar{z}}, z \neq 0.$
5. $z \text{ real} \Leftrightarrow \bar{z} = z$
6. $z \text{ pure imaginary} \Leftrightarrow \bar{z} = -z$
7. $z + \bar{z} = 2a = 2\text{Re}(z)$
8. $z - \bar{z} = 2ib = 2i \cdot \text{Im}(z)$
9. $z\bar{z} = a^2 + b^2$

Geometric representation of a complex number

Let $z = a + ib$ be a complex number, the point $A(a, b)$ is called image or affix of z in the plane. For example:

- $z = 1 - 2i \rightarrow A(1, -2)$
- $A(3, 5) \rightarrow z_A = 3 + 5i$



Geometric representation of $z = a + ib$.

Modulus of complex numbers

Definition

The modulus of complex number $z = a + ib$ is the positive real $|z|$ such that $|z| = \sqrt{z\bar{z}}$ or $|z| = \sqrt{a^2 + b^2}$.

Remark

Let $A(a, b)$ be an affix of $z_A = a + ib$, The length of vector $\overrightarrow{OA}$ is called the modulus of z , where $O(0, 0)$. ($\overrightarrow{OA} = |z_A|$)

In the general case if : $A(a, b)$ and $B(c, d)$ then $AB = |z_B - z_A|$

Examples:

- $z = 3 + 4i \Rightarrow |z| = 5$.
- $z = 3 - 4i \Rightarrow |z| = 5$.
- $z = 1 + i \Rightarrow |z| = \sqrt{2}$.

Properties

1. $|z| = 0 \Leftrightarrow z = 0$
2. $z = z' \Rightarrow |z| = |z'|$, the opposite is not true.
3. $|zz'| = |z||z'|$
4. $|z^2| = |z|^2$
5. $|z^n| = |z|^n, n \in \mathbb{N}^*$
6. $|z + z'| \leq |z| + |z'|$, (The triangular inequality)
7. $\left|\frac{z}{z'}\right| = \frac{|z|}{|z'|}, z' \neq 0$
8. $|z| = |\bar{z}|$

Examples:

- We have $|1 + \sqrt{3}i| = |\sqrt{2} + \sqrt{2}i| = 2$ but $1 + \sqrt{3}i \neq \sqrt{2} + \sqrt{2}i$.
- Let $z \in \mathbb{C}$ such that $|1 + iz| = |1 - iz|$, show that $z \in \mathbb{R}$.

Remarks

- The writing $z = a + ib$ is called the **algebraic form** of the complex number z .
- To write the complex number $\frac{z}{z'}, z' \neq 0$ in algebraic form we use the **conjugate of the denominator**.

Example:

$$z = \frac{1+i}{1+\sqrt{3}i} = \frac{(1+i)(1-\sqrt{3}i)}{(1+\sqrt{3}i)(1-\sqrt{3}i)} = \frac{1-\sqrt{3}i+i+\sqrt{3}}{(1)^2+(\sqrt{3})^2} = \frac{1+\sqrt{3}}{4} + i\frac{1-\sqrt{3}}{4}.$$

Square roots of a complex number

Definition

Let z be a complex number, we say that ω is a **square root** of z , if and only if, $\omega^2 = z$.

Example:

$1+i, -1-i$ are two square roots of $2i$ because $(1+i)^2 = (-1-i)^2 = 2i$.

Proposition

Let z be a complex number, then z admits two square roots $-\omega$ and ω .

Proof.

Let $z = a + ib$ be a complex number, assume that $\omega = x + iy$ a square roots of z , then

$$\omega^2 = z \Leftrightarrow (x + iy)^2 = a + ib$$

$$\Leftrightarrow x^2 - y^2 + i(2xy) = a + ib$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = a \\ 2xy = b \end{cases}$$

Since ω is a square roots of z , so $\omega^2 = z \Rightarrow |\omega^2| = |z|$ that's to say $x^2 + y^2 = \sqrt{a^2 + b^2}$.

Finally, we obtain the following equations system

$$\omega^2 = z \Leftrightarrow \begin{cases} x^2 - y^2 = a & \dots \dots \dots (1) \\ 2xy = b & \dots \dots \dots (2) \\ x^2 + y^2 = \sqrt{a^2 + b^2} & \dots \dots (3) \end{cases}$$

By adding the first and third equations, we get x , then we substitute x into the second equation we get y .

Example:

Find the square roots of the complex number $z = -5 + 12i$.

Second degree equation (Quadratic equations)

Proposition

The second degree equation $az^2 + bz + c = 0$, where $a, b, c \in \mathbb{C}$ and $a \neq 0$, has two solution $z_1, z_2 \in \mathbb{C}$.

Let $\Delta = b^2 - 4ac$ be the discriminant and $\delta \in \mathbb{C}$ a square root of Δ . Then the solutions are

$$z_1 = \frac{-b + \delta}{2a} \text{ and } z_2 = \frac{-b - \delta}{2a}$$

Examples:

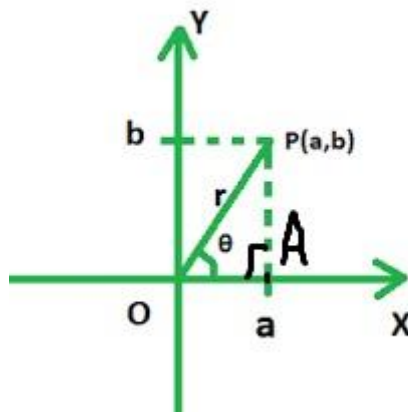
- $z^2 - 4z + 13 = 0$
- $z^2 - (4 + i)z + 5 + 5i = 0$

The trigonometric form of a complex number

Argument of a complex number

Definition

Let $z = a + ib$ be a complex number and let $P(a, b)$ be the image of z in the plane. The angle θ between the vector $\overrightarrow{OP}$ and the axis (OX) is called the **argument** of the number z , we write $\theta = \arg(z)$.



θ : Argument of z .

Remark

From the triangle OAP we have, $\begin{cases} \cos \theta = \frac{OA}{OP} = \frac{a}{r} \\ \sin \theta = \frac{AP}{OP} = \frac{b}{r} \end{cases}$, where $r = |z| = \sqrt{a^2 + b^2}$.

From this remark we have, $a = r \cos \theta$, $b = r \sin \theta$, by substitution in $z = a + ib$ we find that $z = r(\cos \theta + i \sin \theta)$.

Definition

The writing $z = r(\cos \theta + i \sin \theta)$, $r > 0$ is called the **trigonometric form** of the complex number z .

Examples:

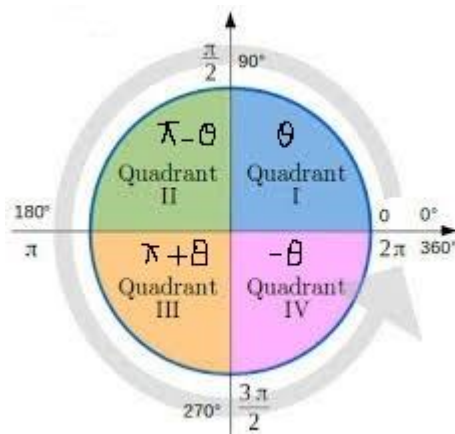
- $z = 1 + i = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$
- $z = 2i = 2 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)$
- $z = -1 = 1(\cos \pi + i \sin \pi)$

Table of some usual angles

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0

- $\cos \pi = -1$
- $\sin \pi = 0$

Quadrant	Sign of a and b	Argument
I (1)	$a > 0$ and $b > 0$	θ
II (2)	$a < 0$ and $b > 0$	$\pi - \theta$
III (3)	$a < 0$ and $b < 0$	$\pi + \theta$
IV (4)	$a > 0$ and $b < 0$	$-\theta$



Circle trigonometry

Example:

$$z = -\sqrt{3} - i$$

$$r = \sqrt{(-\sqrt{3})^2 + (-1)^2} = 2$$

$$\begin{cases} \cos \theta = \frac{-\sqrt{3}}{2} < 0 \\ \sin \theta = \frac{-1}{2} < 0 \end{cases} \Rightarrow \theta = \pi + \frac{\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

$$\theta = \frac{7\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

$$z = 2 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right)$$

Exponential form of a complex number

Definition

Let z be a complex number such that, $|z| = r$ and $\arg(z) = \theta + 2k\pi$.
The writing $z = re^{i\theta}$ is called **Exponential form** of z .

Example:

$$z = 1 + \sqrt{3}i = 2e^{i\frac{\pi}{3}}$$

Proposition

Let z_1, z_2 two complex numbers such as

$z_1 = r_1(\cos \theta_1 + i \sin \theta_1), z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$ then:

$$\triangleright z_1 z_2 = r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)) = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$\triangleright \frac{z_1}{z_2} = \frac{r_1}{r_2} (\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)) = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

$$\triangleright \frac{1}{z_2} = \frac{1}{r_2} (\cos(-\theta_2) + i \sin(-\theta_2)) = \frac{1}{r_2} e^{i(-\theta_2)}$$

$$\triangleright \bar{z}_1 = r_1 (\cos(\theta_1) - i \sin(\theta_1)) = r_1 e^{-i\theta}$$

Moivre's formula

Proposition

Let $z = r(\cos \theta + i \sin \theta) = r e^{i\theta}$ be a complex number, then, for all $n \in \mathbb{N}$;

$$z^n = r^n (\cos n\theta + i \sin n\theta) = r^n e^{in\theta}.$$

Remark

If $z = \cos \theta + i \sin \theta = e^{i\theta}$ then $z^n = \cos n\theta + i \sin n\theta = e^{in\theta}$.

Example:

If $z = 1 + i$ find z^{2024}

We have $z = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$ we apply Moivre's formula we obtain

$$z^{2024} = \sqrt{2}^{2024} \left(\cos 2024 \frac{\pi}{4} + i \sin 2024 \frac{\pi}{4} \right) = 2^{2012} (\cos 506\pi + i \sin 506\pi)$$

$$z^{2024} = 2^{2012}.$$

Remarks

- $\cos 2k\pi = 1, \cos(2k+1)\pi = -1$
- $\sin k\pi = 0$
- $\cos k\pi = (-1)^k, k \in \mathbb{Z}$.
- $\cos \frac{(2k+1)\pi}{2} = 0$
- $\sin \frac{(2k+1)\pi}{2} = 0$

Euler's formulas

If we have: $\begin{cases} e^{i\theta} = \cos \theta + i \sin \theta \\ e^{-i\theta} = \cos \theta - i \sin \theta \end{cases}$, then for all $\theta \in \mathbb{R}$:

$$\begin{cases} \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \\ \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \end{cases}$$

Example:

Write in linear form $(\cos \theta)^2$ and $(\sin \theta)^2$

We have

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \Rightarrow (\cos \theta)^2 = \left(\frac{e^{i\theta} + e^{-i\theta}}{2} \right)^2 = \frac{e^{2i\theta} + e^{-2i\theta} + 2}{4} = \frac{2 \cos 2\theta + 2}{4}$$

$$(\cos \theta)^2 = \frac{1}{2} + \frac{1}{2} \cos 2\theta.$$

Similarly we find

$$(\sin \theta)^2 = \frac{1}{2} - \frac{1}{2} \cos 2\theta.$$

nth roots of a complex number

Definition

For $z \in \mathbb{C}$ and $n \in \mathbb{N}$, a **nth root** is a number $\omega \in \mathbb{C}$ such that $\omega^n = z$.

Proposition

There are n n-th roots $\omega_1, \omega_2, \dots, \omega_{n-1}$ of $z = \rho e^{i\theta}$, these are:

$$\omega_k = \rho^{\frac{1}{n}} e^{\frac{i\theta + i2k\pi}{n}}, k = 0, 1, \dots, n-1.$$

Example:

Solve the equation: $z^5 = 1$.

Assume that $z = r e^{i\theta}$, on the other hand we have $1 = 1 e^{i(0)}$

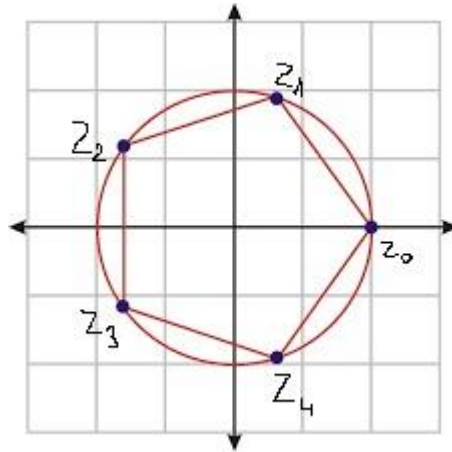
$z^5 = 1 \Rightarrow r^5 e^{i5\theta} = 1 e^{i(0)}$ by identification we find

$$\begin{cases} r^5 = 1 \\ 5\theta = 0 + 2k\pi \end{cases} \Rightarrow \begin{cases} r = 1 \\ \theta = \frac{2k\pi}{5} \end{cases}$$

then $z_k = e^{\frac{2ki\pi}{5}}, k = 0, 1, 2, \dots, 4$.

So the solutions of the equation $z^5 = 1$ are:

$z_0 = 1$	$z_1 = e^{\frac{2i\pi}{5}}$	$z_2 = e^{\frac{4i\pi}{5}}$	$z_3 = e^{\frac{6i\pi}{5}}$	$z_4 = e^{\frac{8i\pi}{5}}$
-----------	-----------------------------	-----------------------------	-----------------------------	-----------------------------



5th roots of 1.